

Research Paper Summary: Paper-Based Coding Exams & CodeShuffler

Paper: “Paper-based coding exams: Overcoming coding assessments and grading challenges in the era of AI Large Language Models like ChatGPT”

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1. Short Summary

The research paper studies the difficulty of fairly assessing students’ real coding ability when AI tools such as ChatGPT can generate programming solutions. The authors compare computer-based auto-graded coding assignments with paper-based coding assessments. They propose a paper-based assessment strategy supported by a software tool called **CodeShuffler**. CodeShuffler takes correct code, shuffles its lines, numbers them, and generates multiple-choice options containing different line sequences. Students must understand the code and identify or arrange the correct sequence.

2. Main Problem

- Students can use AI language models to generate code for online assignments.
- A technically correct submission may not represent the student’s actual programming knowledge.
- Traditional plagiarism detection may not reliably identify AI-generated code.
- Manual grading of paper-based coding exams can be time-consuming, especially in large classes.

3. Proposed Solution — CodeShuffler

The authors developed CodeShuffler to reduce the grading burden while keeping the assessment paper-based. The instructor supplies the correct code. The tool shuffles the code lines, assigns line numbers, creates four customizable answer choices, and identifies the correct sequence. The shuffled code can also be produced as an image for inclusion in an exam paper.

4. Technologies / Tools Used

Technology / Tool	Purpose
Python	Programming language used in the introductory programming course studied.
CodeShuffler	Shuffles code lines and generates multiple-choice sequences.
zyBooks	Online auto-graded coding platform used for comparison.
ZipGrade	Can be used to check paper-based MCQ answers.
GradeCam	Can be used for automated answer checking.
Gradescope	Can be used for grading/assessment.
ChatGPT	Used in the study to examine AI-generated code and compare solutions.

5. Advantages

1. Reduces the impact of AI-generated ready-made solutions during the assessment.
2. Tests students’ understanding of code structure and sequencing.
3. Reduces instructors’ grading workload through MCQ-based checking.

4. Can be useful for large-enrollment programming courses.
5. The approach is not limited to Python; the paper reports use with other programming languages, including assembly.
6. Providing code lines as a reference can help students organize their solution and reduce exam anxiety.

6. Disadvantages / Limitations

1. The paper-based format does not fully test typing, program execution, debugging, or other practical coding activities.
2. Preparing the paper-based assessment still requires instructor effort.
3. Multiple-choice formats can allow some possibility of guessing.
4. Very large or highly complex programs may be difficult to represent conveniently as shuffled lines.
5. Compared with an online IDE, students do not get immediate execution and debugging feedback.

Note: The paper does not provide a separate section explicitly listing these disadvantages; the points above are limitations that follow from the assessment method described in the paper.

7. Experiment and Results

The authors first used a paper-based coding problem to examine students' fundamental understanding. They later tested the proposed approach with a larger sample of **223 students**. The study reports that **73% of students** began their coding process on a rough sheet. The correlation between paper-based and computer-based coding scores was **0.42** after the mid-semester assessment and **0.71** after the final exam, showing a stronger relationship in the final assessment.

8. Simple Working Flow

1. Instructor prepares the correct code.
2. CodeShuffler shuffles the code lines.
3. The tool assigns line numbers and creates four possible sequences.
4. The shuffled code is added to the exam paper.
5. Student arranges/identifies the correct sequence.
6. Student selects the correct MCQ option.
7. Answers can be checked using an MCQ grading tool.

9. Presentation-Ready Explanation

"This research paper focuses on the problem of evaluating students' actual coding skills in the era of AI tools like ChatGPT. The authors propose a paper-based coding assessment supported by CodeShuffler. CodeShuffler shuffles code lines and generates multiple-choice options, so students have to understand and arrange the code correctly. This reduces dependence on AI-generated solutions during assessment and also reduces the instructor's grading workload."

10. Conclusion

The paper concludes that paper-based coding assessments supported by CodeShuffler can provide a practical way to assess fundamental coding and code-structuring skills while reducing grading effort. The tool can generate shuffled code and multiple-choice options, and it can optionally introduce incorrect code lines to increase assessment difficulty. The authors state that CodeShuffler and the assessment template

are open-source.

Source: Baig, Hasan & Bradford, Phillip, 2025 ASEE Northeast Section Conference paper. This summary is based on the uploaded research paper.